

$$\begin{array}{r} M = 1000100 \\ N = 1010100 \end{array}$$

$$2\text{'s complement of } N = 0101011$$

$$\begin{array}{r} 1000100 \\ + 0101011 \\ \hline 1110000 \rightarrow \text{no carry.} \end{array}$$

$$\begin{array}{r} 0101100 \\ 2\text{'s complement of } 1110000 : \\ 0001111 \\ \hline 0010000 \end{array}$$

answer: $-(10000)$

Proof of the procedure.

The addition of M to the r 's complement of N gives $(M + r^n - N)$. For numbers having an integer part of n digits, r^n is equal to a 1 in the $(n+1)$ th position (which has been called the "end carry"). Since both M and N are assumed to be positive, then:

- (a) $(M + r^n - N) \geq r^n$ if $M \geq N$, or
 (b) $(M + r^n - N) < r^n$ if $M < N$.

In case (a) the answer is positive and equal to $M - N$, which is directly obtained by discarding the end carry r^n . In case (b) the answer is negative, and equal to $-(N - M)$. This case is detected from the absence of an end carry. The answer is obtained by taking a second complement and adding a negative sign:

$$- [r^n - (M + r^n - N)] = -(N - M).$$

1.5.4 Subtraction with $(r-1)$'s Complement

The procedure for the subtraction with the $(r-1)$'s complement is exactly the same as the one used with the r 's complement except for one variation, called "end-around carry". The subtraction of $M - N$, both positive numbers in base r , may be calculated in the following manner:

1. Add the minuend M to the $(r-1)$'s complement of the subtrahend N .
2. Inspect the result obtained in step 1 for an end carry.

(a) If an end-carry occurs, add 1 to the least-significant digit (end-around carry).

(b) If an end-carry does not occur, take the $(r-1)$'s complement of the no. obtained in step 1 and place a negative sign in front.